



UNIVERSITY OF EASTERN AFRICA, BARATON

SCHOOL OF BUSINESS

DEPARTMENT OF INFORMATION SYSTEM AND
COMPUTING

SENIOR PROJECT WRITTEN IN PARTIAL
FULFILLMENT OF THE REQUIREMENT OF THE
COURSE: COSC492

COURSE TITLE: SENIOR PROJECT

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SENIOR PROJECT

TOPIC: AUTOMATION OF LESSON PLAN, SCHEMES OF WORK AND RECORD OF WORK FOR STUDENTS.

PROBLEM STATEMENT

Students spend a lot of time manually preparing lesson plans, schemes of work and record of work. This manual process is often repetitive, time consuming and prone to writing errors leading to delay in lesson delivery and documentation. Physical documentation can be easily lost and be difficult to retrieve and update. This affects negatively to lesson continuity and student progress tracking.

Students on attachment often struggle with maintaining a well structured lesson plan, schemes of work and records of work due to the manual and paper based nature of this processes. Supervisors and lectures face challenges in monitoring students progress, verifying lesson coverage and ensuring adherence to the required training curriculum. The lack of a centralized automated system leads to inefficiencies, inconsistent reporting and difficulties in tracking students performance during the attachment period.

PROPOSED SOLUTION

An automated system for managing lesson plans, schemes of work and records of work would improve accessibility and enhance accuracy by digitizing and centralizing these records which can save time for educators and ensure consistency in curriculum planning, facilitate real-time updates of students progress and supervisors can remotely supervise their work in real time, institutions can also ensure standardization in training of their students. This solution would also provide analytical insights into lesson coverage, students progress and curriculum adherence improving the quality of education delivery. This solution would also improve accountability , reduce paperwork and provide data driven insights to enhance the attachment experience and training outcomes.

OBJECTIVES

- 1. To develop a system that automate and digitize the creation, updating and management of lesson plans, schemes of work and records of work. It will also reduce manual workload, minimize errors and improve documentation consistency.**
- 2. To improve monitoring and tracking-** the system will provide a centralized platform for supervisors and lecturers to remotely track students progress, review lesson coverage and

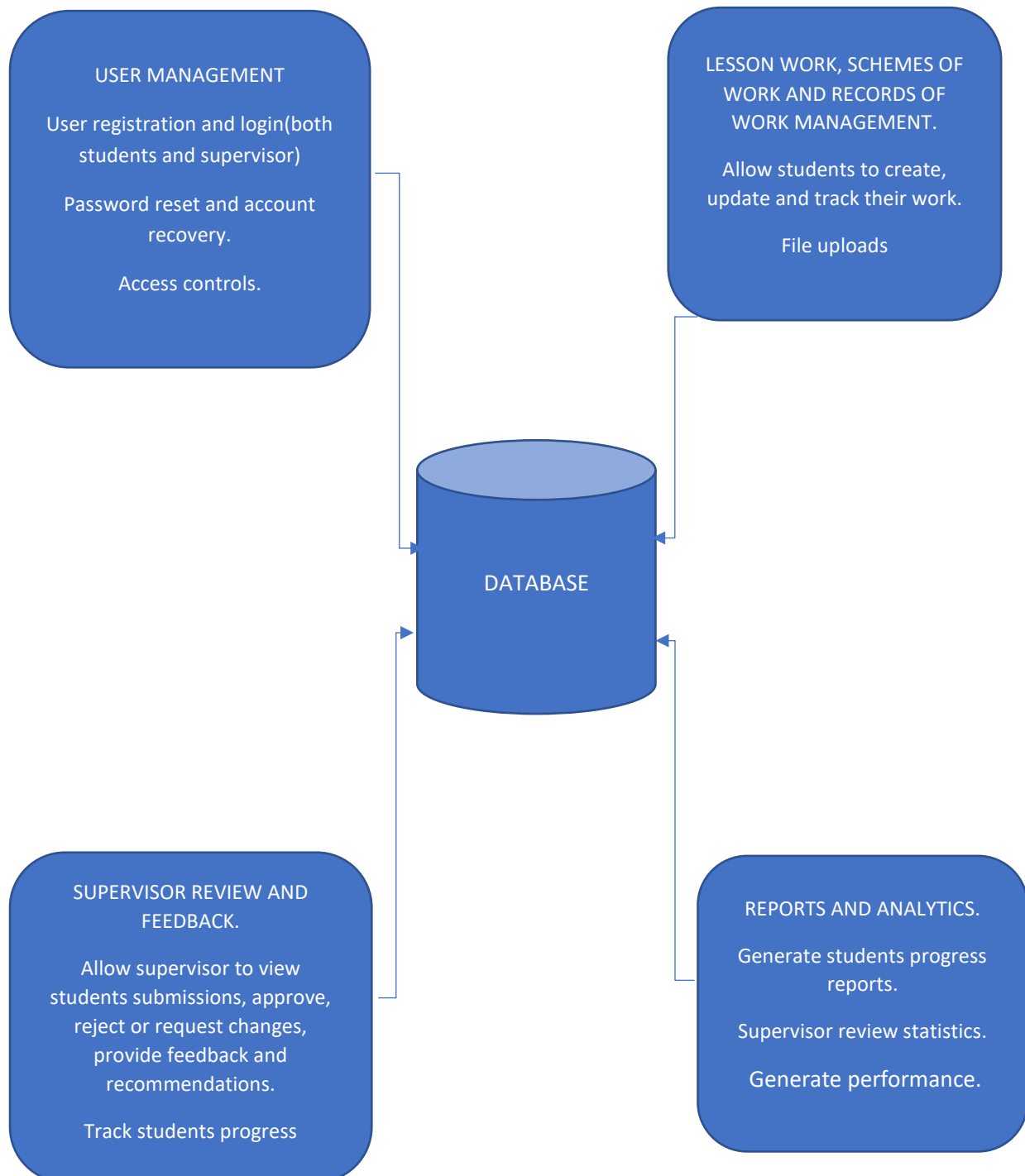
ensure adherence to the required curriculum during the attachment period. The tracking of students during attachment can reduce their probability of being lazy.

3. **Ensure Accessibility and standardization-** The system will enable students, supervisors and institutions to access and update records in real time from any location ensuring uniformity, easy retrieval and better collaboration between stakeholders.

TOOLS AND SOFTWARES

1. Programming languages – Python, JavaScript.
2. Database – MySQL.
3. Cloud services – Firebase for real-time data management.
4. Frameworks – Django for backend development and React for frontend.
5. Testing and debugging – postman for API testing.
6. Version control and collaboration – GitHub for code management and version control.

CONCEPTUAL DIAGRAM



Senior Project : Grading Template

Date:

Student Name:

Project Title:

Criteria:

- | | | |
|---|------------------------------------|---------------|
| 1. | Project Proposal (10%) | Marks Awarded |
| <ul style="list-style-type: none">· Clarity of project objectives and scope· Relevance and significance of the problem or challenge addressed· Feasibility of proposed solution approach· Adequacy of project plan and timeline | | |
| 2. | Project Plan (15%) | Marks Awarded |
| <ul style="list-style-type: none">· Comprehensive project plan outlining tasks, milestones, and dependencies· Clear allocation of resources (e.g., personnel, budget, equipment)· Realistic timeline with achievable deadlines· Identification and mitigation of potential risks | | |
| 3. | Midterm Progress Report (15%) | Marks Awarded |
| <ul style="list-style-type: none">· Progress made towards project objectives and milestones· Adherence to project plan and timeline· Identification and resolution of any challenges or issues encountered· Quality of documentation and reporting | | |
| 4. | Final Project Implementation (30%) | Marks Awarded |
| <ul style="list-style-type: none">· Completion of project deliverables as outlined in the project plan· Functionality and effectiveness of the ICT solution· Quality of code, software, or hardware developed· User interface design and usability | | |
| 5. | Final Project Report (15%) | Marks Awarded |
| <ul style="list-style-type: none">· Comprehensive documentation of project process, methodology, and outcomes· Clear presentation of findings, results, and conclusions· Organization, clarity, and coherence of written report | | |

- Proper citation of sources and references
- 6. Presentation and Defense (15%) Marks Awarded
 - Clarity and effectiveness of oral presentation
 - Ability to communicate project objectives, methodologies, and outcomes to diverse audiences
 - Response to questions, feedback, and challenges during the defense
 - Professionalism, confidence, and engagement during the presentation
- 7. Overall Evaluation (10%) Marks Awarded
 - Overall quality and completeness of the senior project
 - Demonstrated mastery of ICT concepts, principles, and skills
 - Contribution to the advancement of knowledge or practice in the field
 - Reflection on personal learning, growth, and areas for improvement

Total Grade:

Additional Comments: